2025_Hexaware_Actual_Test_1

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

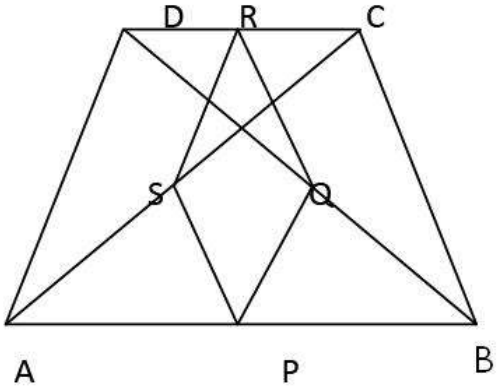
Additional Instructions:

None

Q1. The mean temperature from Monday to Wednesday was 37°C and from Tuesday to Thursday was 34°C, if the temperature on Thursday was 4/5th that of Monday, then what was the temperature on Thursday?

- 36
- 36.5
- 35.5
- 34

Q2. ABCD is trapezium in which AB||DC and AD = BC. IF P,Q,R,S be respectively the mid-points of BA, BD and CD, CA. The PQRS is



- Rhombus
- rectangle
- parallelogram
- square

Q3. If $3^{(X-2)} = 5$ and $\log_{10} 2 = 0.30103$, $\log_{10} 3 = 0.4771$, then X =

1 22187/47710

2 22187/47710

3 22187/47710

None of these

Q4. If the measures of one side and one diagonal of a rhombus are 13 cm and 24 cm respectively then its area in cm^2 is:

72

84

80

120

Q5. Two clocks chime once every P minutes and every Q minutes respectively. If P and Q are primes then at what intervals do the two clocks chime together?

P + Q min

$\frac{P}{Q}$ min

PQ min

$P^2 + Q^2$ min

Q6. What is the rightmost non-zero digit in 15!?

8

10

15

14

Q7. In a college of 1335 students, 60% are boys. Find the number of girls and the number of boys in the college.

924, 221

691, 404

802, 333

534, 801

Q8. A box contains 2 white balls,3 black balls, and 4 red balls.In how many ways can 3 balls be drawn from the box,if at least one black ball is to be included in the draw?

64

32

48

96

Q9. A polygon has 54 diagonals. Find the numbers of its sides.

6

12

27

18

Q10. A room contains 3 red, 5 green, and 4 blue chairs. Two chairs are picked and put on the lawn. What is the probability that none of the chairs picked is blue?

5/33

9/33

14/33

17/33

Q11. A man sells an article at a loss of 20%. Had he bought it for 20% less and sold it for Rs. 60 more, he could have gained 50%. What is the cost price of the article?

Rs.150

Rs.100

Rs.200

Rs.120

Q12. A girl buys 2 pigeons for ₹182. She sells one at a loss of 5% and another at a profit of 8%. But she neither gains nor loses on the whole. Find the price of a pigeon that has sold at a profit.

Rs. 112

Rs. 70

Rs. 85

cant be determined

Q13. The sum of all the terms of the A.P. 7, 10, 13, ...N is 1242, where N is the last term of the A.P. Find the value of N.

67

79

85

102

Q14. Two numbers are such that the ratio between them is 3 : 4. But if each is increased by 15, the ratio between them becomes 4 : 5. The bigger number is

45

50

55

60

Q15. Six litres of milk were taken out from a vessel and is filled with water. This operation is performed two more times. The ratio of the quantity of milk now left in the vessel to water is 8:19. How much is the quantity of milk contained by the vessel originally?

18 L

16 L

14 L

12 L

Q16. If the difference between CI and SI earned on a certain amount at 20% pa at the end of 3 years is Rs.640, find out the principal.

Rs5500

Rs6500

Rs4500

Rs5000

Q17. The rates of interest offered by the two banks BOI and SBI are 8% and 44% per annum at compound interest. Mohan deposits certain but different sums of money in both banks. After three years, he received equal amounts from both the banks. The ratio of amounts invested in BOI and SBI respectively are:

44 : 8

64 : 8

125 : 64

64 : 27

None of these

Q18. A can do a piece of work in 6 days for which B takes 8 days. C takes as long as A and B would take working together. How long will it take B and C to complete the work together?

25/7 days

12/5 days

24/7 days

27/4 days

Q19. In a 750 m race, the ratio of speeds of Arun to Darani is 1:3. Arun has a head start of 250 m. Who wins the race and by what distance?

Arun, 200 m.

Darani, 250 m.

Arun, 250 m.

Darani, 200 m.

Q20. What should be the value of a, in the polynomials x^2-11x+a and x^2-14x+2a, so that these two polynomials have common factors?

24

1

-1

1/2

Q21. **Complete the following sentence by choosing the right option.**

The board meeting was called ..... due to the sudden demise of the chief executive officer.

for

off

to

out

Q22. The question given below is followed by two arguments numbered I and II. You have to decide which of the arguments is a 'strong' argument and which is a 'weak' argument.

Statement :
Is paying ransom or agreeing to the conditions of kidnappers of political figures, a proper course of action?

Arguments :
I. Yes. The victims must be saved at all costs.
II. No. It encourages the kidnappers to continue their sinister activities.

if only argument I is strong

if only argument II is strong

if neither I nor II is strong

if both I and II are strong

Q23. In the following question, statements 1 and 6 are respectively the first and the last sentences of a paragraph and statements A, B, C, and D come in between them. Rearrange A, B, C, and D in such a way that they make a coherent paragraph together with statements 1 and 6. Select the correct order from the given choices and mark its number as your answer.

1. Very popular with the younger crowd of the twin cities are bowling alleys.

A. They are not allowed to use the bowling alley.

B. But tough luck for those less than 12 years.

C. Their plush interiors, music, and cafeteria facilities make them the ideal places for youngsters to take their friends out for a treat.

D. There are already four of them in town with another ready to join the club.

6. The older kids who use them swear they have the best time in the world.

- CBAD
- ABCD
- DCBA
- BADC

Q24. Fill in the blank with the right discourse marker:

The interiors are beautiful and _____, they have designed them themselves.

- obviously
- by the way
- what's more
- None of the these

Q25. Complete the following sentences by selecting the appropriate word from the given choices.

Dowry is no longer permitted by law even in_____ marriages.

- natural
- conventional
- polygamous
- love

Q26. Read each sentence to find out whether there is any error in it. The error, if any, will be in one part of the sentence. The number of that part is the answer. Mark "No error" if the sentence is correct.

(1) Injustice and / (2) discrimination can / (3) never be tolerated / (4) by no one.

- 1
- 2

3

4

No error

Q27. In the following sentence, a part of the sentence or the whole sentence is underlined. Beneath each sentence, four different ways of phrasing the underlined part are indicated. Choose the best alternative from among the four.

Federal incentives now encourage investing capital in commercial office buildings despite vacancy rates in existing structures that are exceptionally high and no demand for new construction.

capital investment in commercial office buildings, even though vacancy rates in existing structures are exceptionally high and there is

capital to be invested in commercial office buildings even though there are exceptionally high vacancy rates high in existing structures with

investing capital in commercial office buildings even though the vacancy rates are exceptionally high in existing structures with

capital investment in commercial office buildings despite vacancy rates in existing structures that are exceptionally high, and although there is

Q28. **Convert from direct to indirect speech**

The woman said, “No, I refuse to confess guilt.”

The woman emphatically refused to confess guilt.

The woman refused to confess her guilt.

The woman told that she did not confess guilt.

The woman was stubborn enough to confess guilt.

Q29. Identify the synonyms of the capitalized words

TREACHERY

artifact

novelty

betrayal

charm

Q30. Choose the word that is opposite in meaning to the given words.

Capricious

- thoughtful
- specious
- constant
- capacious

Q31. Substitute the given sentences with one word.

Hater of learning and knowledge

- Misologist
- Bibliophile
- Misogynist
- Misanthropist

Q32. Change the voice for the given sentence.

He had kept up his diet program for a month.

- His diet program had been kept up for a month.
- His diet program was kept up for a month.
- His diet program had been kept for a month.
- The doctor had kept up his diet program for a month.

Q33. **Find the missing term:**

APK, BQL, CRM, DSN, ?

- TEO
- ETO
- EOT

TOE

Q34. A man pointing to a photograph says, “The lady in the photograph is my nephew’s maternal grandmother”. How is the lady in the photograph related to the man’s sister who has no other sister?

Mother

Sister

Sister-in-law

Mother-in-law

Q35. What was the day on 14th Mar 1993?

Friday

Thursday

Sunday

Saturday

Q36. What is the angle covered by the minute's hand in 20 minutes?

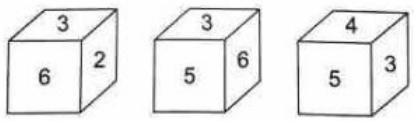
120°

130°

140°

150°

Q37. Three positions of a dice are given below
Identify the number on face opposite of 6?



4

1

5

7

Q38. The following problem consists of a question and three statements I, II, and III given below. Read all the statements carefully and seek all possible combinations which could be sufficient for answering a question.

Who among Girish, Prashant, Kamlesh, and Jayesh is the youngest?
I. Kamlesh Is younger than Jayesh but older than Girish and Prashant.
II. Jayesh is the oldest.
III. Girish is older than Prashant.

I and II only

I and III only

II and III only

I only

Q39. The question below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both statements and give an answer.

If Jai's and Kishore's annual salaries in 1985 were each 10 percent higher than their respective annual salaries in 1984, what was Jai's annual salary in 1984?
(A) The sum of Jai's and Kishore's annual salaries in 1984 was Rs.50,000.
(B) The sum of Jai's and Kishore's annual salaries in 1985 was Rs.55,000.

if the question can be answered by using statement A alone but not by using B alone.

if the question can be answered by using statement B alone but not by using A alone.

if the question can be answered by using either statement alone.

if both the statements taken together are not sufficient and more information is required to solve the question

Q40. Amit starts from his house, walks 4 km North, took a right turn walked 4 km, from there he took a left turn and walked 4 km. From there, his school is 2 km to his right. How far is his house and in what direction from school?

10 km North-west

10 km North-east

10 km South-west

6 km West

Q41. What will come in place of question mark (?) in the given number series?

3, 72, 129, 173, 203,?

227

254

252

218

Q42. **In the question below are given three statements followed by four conclusions numbered I, II, III and IV. You have to take the given statements to be true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follows from the statements, disregarding commonly known facts. Give answer.**

Statements:
All Gold are Platinum.
No Platinum is silver.
Some Diamonds are silver.

Conclusion:
I. Some Diamonds are Gold
II. Some Diamonds are Platinum
III. Some Gold are Silver
IV. No Silver is Gold

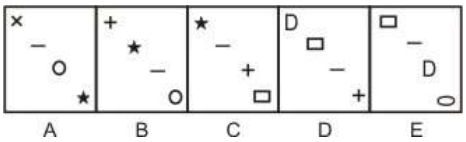
Only I follows

Only III follows

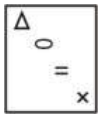
Only IV follows

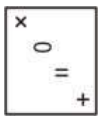
Only II and IV follow

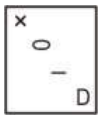
Q43. Find the next Figure.











Common Content:

Fill in the appropriate words to complete the passage.

Clement Atlee became the Prime Minister of England after the Second World War. Winston Churchill who had successfully..(1)..England and the allies to victory over Hitler was now rejected by the English people at the hustings. Labour Party was..(2)..to power and Atlee became the Prime Minister. One of his memorable tasks was that he was..(3)..in granting India its freedom. Atlee was born in a well-to-do..(4)..but he always had..(5)..for the poor and the down-trodden. He is known for keeping..(6)..and cooperation among his cabinet colleagues. Yiot that there were .no differences of opinion..(7)..his cabinet members, but Atlee, by his..(8)..nature and positive approach, always managed to keep them together and had control over them..(9)..being sympathetic to the cause of India, and granting India freedom, he..(10)..many a constructive activity for his country too, like nationalization of some industries, and starting national health scheme.

Q44. Choose the appropriate word for the blank numbered(2)

- averse
- swept
- immune
- apathy

Q45. Choose the appropriate word for the blank numbered(1)

- isolated
- conquered
- led
- marginalised

Q46. Choose the appropriate word for the blank numbered(3)

- interested
- instrumental
- eager
- reluctant

Common Content:

Read the passage and answer the following questions:

What is common between Pope Paul III, former Prime Minister Morarji Desai, Italian composer Goeehina Rossini, Gone with the Wind, Playboy Club and Pulsar CP1919 (a star that sends signals every 1.337301 seconds)?

February 29, is the extra day that comes every four years. While the Pope, Rossini and Morarji Desai were born on this date, the movie Gone with the Wind won eight Oscars that day in 1940, the first Playboy Club featuring bunnies opened in Chicago (1960), and CP1919, the first pulsar was discovered (1968).

Ever wondered why the extra day occurs every four years? It is to keep the seasons in their right months. It takes the earth slightly more than 365 days - about 5 hours 48 minutes and 45.22 seconds - to, complete one rotation around the sun. Multiply that into four years and it would be one day extra. After four years the earth would be short of one day to make four complete rotations around the sun, and, in 120 years, a full month. In 120 years, the seasons would come a month later. This is why we give back a day to nature by inserting a day in the calendar every fourth year.

Emperor Julius Caesar introduced the system of leap year in 46 BC, it was found that the leap year was introduced by the priests every third year instead of the fourth and, to compensate for this error, no more leap years were added until 8 AD. But by giving one day back to nature the year had 11 minutes and about 14 and three-fourth seconds extra. By the 15th century, this added up to 10 extra days. Pope Gregory XIII decided to make the 10-day correction all at once. He shortened October 1582 by 10 days: October 4, 1582, was followed by October 15, 1582. The rule for deciding the leap year was also modified. If a century is divisible completely by 400 only then it would be a leap year. The year 1900 was not a leap year but 2000 was. This calendar is, however, not perfect: an error of three days crops up after 10,000 years.

Q47. The extra day is added to the leap year

- as Pope Paul decreed so.
- to keep the seasons in their right months.
- because earth's rotation is erratic.
- since the sun also moves in space.

Q48. The system of leap year was introduced by

- Julius Caesar.
- Rossini.
- Pope Gregory XIII.
- Pope Paul III.

Q49. The calendar set by Pope Gregory XIII was still not perfect because

- expertise was lacking.
- of the disagreement among the people regarding leap year
- an error of three days occurs after 10,000 years.
- it lacked scientific calculations.

Q50. Pope Gregory XIII

- decided that 1900 would not be a leap year.
- decided that 2.000 would be a leap year.
- made a 10 day correction in the calculations.

decided to modify the rule for calculating a leap year.

Q51. The phrase 'to give back a day to nature' means

to omit a day in the calendar.

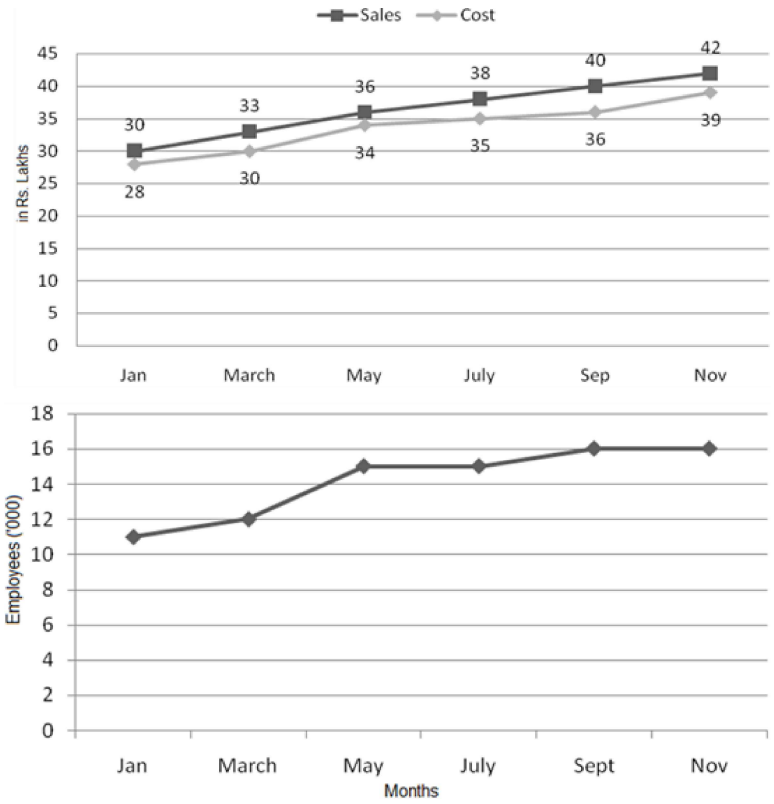
to add a day to make the calendar conform to the seasons.

to spend a day in observing natural phenomena.

to care for nature at least one day in a year.

Common Content:

Refer to the graphs to answer the questions that follow.
The following line graphs show the data of the company. The first line graph shows the sales and cost which is Rs. lakhs and the below line graph shows the number of employees (in thousands).



Q52. Which one of the following months records the highest profit?

January

March

September

November

Q53. The average profit for the period is

2.57 lakhs

3.63 lakhs

2.83 lakhs

3.12 lakhs

Q54. Which month has the lowest profit per employee?

January

May

September

November

Q55. Assuming that no employee left the job, how many more people did the company take on in the given period?

2000

3000

4000

5000

Common Content:

Study the following information carefully and answer the questions given beside:

Eight persons – Navya, Farukh, Ankur, Mangal, Ravan, Danav, Simar, and Gopal are seated around a circular table such that only three of them are facing away from the centre and the rest are facing towards the centre (not necessarily in the same order).

Navya sits second to the left of Ankur. Farukh is not an immediate neighbour of both Ankur and Navya. One of the immediate neighbours of Ankur faces the opposite direction of Ankur. Gopal sits third to the left of Navya. Danav sits second to the left of Gopal. Mangal is not an immediate neighbour of Farukh. Simar sits third to the left of Mangal. Ravan and Farukh face the same direction as Mangal faces. Farukh doesn't sit opposite to both Ankur and Navya.

Q56. Who among the following sits second to the right of Danav?

Navya

Simar

Mangal

Gopal

Q57. Who among the following sits third to the left of Simar?

- Mangal
- Danav
- Navya
- Ravan

Q58. How many person(s) sit between Ravan and Navya when counted from the right of Navya?

- One
- Two
- Three
- Four

Q59. Who among the sits in front of Ravan?

- Farukh
- Mangal
- Navya
- Ankur

Q60. If the all persons are made to according to their names appear in dictionary in clockwise direction starting from Ankur then position of how many person(s) will remain unchanged including Ankur?

- One
- Two
- Three
- Four

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

[Note: If (X) gets executed if the value inside if( ), ie, x is not zero].

```
1  integer a, b, c
2  set a=3, b=2
3  if (b-1)
4  for (each c from 1 to 3)
5  a= a-1
6  end for
7  for(each c from 1 to 3)
8  a= a + b
9  end for
10 end if
11 print a+ b
```

- 1
- 8
- 4
- 5

Q2. What will be the output of the following pseudocode?

[Note:&&: Logical AND - The logical AND operator (&&) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0)].

| |: Logical OR - The logical OR operator (| |) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0), otherwise.

```
1  integer a1, a2, p
2  set a1= 16, a2= 234
3  p= (a1 + a2) AND (a1) OR (a2)
4  if ( p not equqls 0)
5  print p + p
6  else
7  print p
8  end if
```

- 500
- 1
- 4
- 3

Q3. What will be the output of the following pseudocode?

```
1  integer a, b, j
2  set a=10, b= 18
3  for (i=6 to a-1) increase i by
4  b= ((b*2)/ 2) + 3
5  end for
6  print b
```

21

24

27

30

Q4. What will be the output of the following pseudocode?

```
1  integer a, b, c
2  set a=13, b= 9, c= 6
3  if (a< b)
4  if (a < c)
5  print a
6  else
7  print c
8  end if
9  else if (b < c)
10 print b
11 else
12 print c
13 end if
```

6 9

9

6

13

Q5. What will be the output of the following pseudocode?

[&& : Logical AND - The logical AND operator (&&) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0)
If (X) gets executed if the value inside if( ), ie, x is not zero].

```
1  integer i, a, j
2  set i=6, a=23
3  j= i + a && i / a && a/ i + i
4  if (j)
5  print j
6  else
7  print j + j
8  end if
```

0

20

2

1

Q6. What will be the output of the following pseudocode?

```
integer a, b , a, d
set a=3, b= 1, c=1
for (each d from 1 to 3)
a= a-1
if(d> 2)
a= a- b
b=a
if (d < 2)
jump out of the loop
end if
a= a + 1
end for
print a + c
```

3

-5

4

5

Q7. What will be the output of the following pseudocode?

```
integer a, b
for (each a from 1 to 2)
b= 17
print b + a - a * a
end for
print a + 1
```

17 17 16

17 15 4

1 2 3

17 4 15

Q8. What will be the output of the following pseudocode?

```
integer k
set k=6/7 + 3 - 3
if ( k < = 0)
if ( k equals 0)
print "test1"
else
print " test2"
end if
else
print "test3"
end if
print " test completed"
```

test3 test completed

test2 test completed

test completed

test1 test completed

Q9. What will be the output of the following pseudocode?

```
integer j, m
set m=1
integer a[4]= {0, 1,0,2}
a[0] = a[0] - a[2] + a[3]
a[1] = a[0] -a[3]
a[2]= a[1] - a[2]
a[3] = a[2]
m = a[3]
print m
```

6

0

2

1

Q10. What will be the output of the following pseudocode?

```
integer n
set n=15
while( n <=15)
n= n + 6/3
print "biography"
continue with next iteration
end while
```

it will print biography 13 times

it will print biography 15 times

it will print biography 1 time

it will print biography 14 times

Q11. What will be the output of the following pseudocode for p= 8?

```
integer combine(integer p)
if (p < = 17)
combine(combine(p * 2))
end if
return p + 3
end function combine ()
```

13

34

41

12

Q12. What will be the output of the following pseudocode for a= 22?

```
1 integer compact(integer a)
2 if( a <= 45)
3 compact( compact (compact (a * 5 + 1)))
4 end if
5 return a*2
6 end function compact ()
```

22

45

56

888

Q13. What will be the output of the following pseudocode if a = 10?

```
1 integer f5(integer a)
2 if (a mod 2)
3 return a++
4 else
5 return f5(f5 ( a- 1))
6 end if
7 end function f5()
```

9

98

101

87

Q14. What will be the output of the following pseudocode if a = 2 b = 7?

[Note:If (X) gets executed if the value inside

if( ), ie, x is not zero].

```
1  integer funn( integer a, integer b)
2  if (a > -2)
3  if(b + 1)
4  return 2
5  end if
6  return a
7  end if
8  return b
9  end function funn ( )
```

- 0
- 2
- 11
- 1

Q15. What will be the output of the following pseudocode?

[Note: MOD finds the remainder after the division of one number by another. For example, the expression "5 MOD" would evaluate to 1 because 2 leaves a quotient of 2 and a remainder 1].

```
1  character a[20], b[20]
2  set a[20] = "Alpital"
3  integer j, k
4  set j= 0, k=0
5  while (a[j] NOT EQUALS null character)
6  if (j mod 2 EQUALS 0)
7  b[k]= a[j]
8  else
9  b[k]= ' '
10 end if
11 k= k + 1
12 j= j + 1
13 end while
14 set k= 0
15 while (b [k] NOT EQUALS null character)
16 print b [k ]
17 k= k + 1
18 end while
```

- t a l p
- l l a
- A t p l
- A p t l

Q16. All the options mentioned below are real security and privacy risks, except:

- Virus attack
- Spam

Worm attack

Malware attack

Q17. Consider the entities A1 and A2 in an ER diagram have simple single-valued attributes. B1 and B2 are the relations between A1 and A2. B1 is a one-to-many and B2 is a many-to-many relationship. There are no attributes present in B1 and B2. How many tables are required to represent this given model?

3

2

5

4

Q18. Match the following logical functions with their use.

1. IF
2. NOT
3. AND

A – Return TRUE if all of its arguments are TRUE.
B – Specifies a logical test to perform
C – Reverses the logic of its argument

1-A, 2-C, 3-B

1-B, 2-A, 3-C

1-B, 2-C, 3-A

1-A, 2-B, 3-C

Q19. Which of the following options will you use if you want a person to be informed of the mail but not exactly the intended receipt and you want everyone to explicitly see all the recipients?

Bcc

Cc

Post

Mail Merge

Q20. While sending e-mails, which option should be chosen if you want to put another person in a copy secretly?

Bcc

To

Subject

Cc

Q21. Which of the following should be used to store the files securely and open or edit them from any device having an internet connection?

Google edit

Adobe Acrobat

All of the mentioned options

Google drive

Q22. Classify the following as Input or Output devices.

1. Keyboard
2. Mouse
3. Printer
4. Monitor

1->Input device
2->Input device
3->Output device
4->Output device

1->Input device
2->Input device
3->Output device
4->Input device

1->Input device
2->Output device
3->Output device
4->Input device

1->Input device
2->Output device
3->Input device
4->Output device

Q23. What could be the possible reason for getting the given error in a worksheet?
#NUM!

Text not recognized in a formula

A value is not available to a function

Function contains invalid numeric values

A number is divided by zero

Q24. Which of the following tasks can be done with the Internet?

1. Media download
2. File transfer
3. Financial Transaction

Choose the correct answer from the options given below.

Only 1 and 2

Only 1 and 3

Only 2 and 3

All 1, 2 and 3

Q25. Match the following file types in List 1 with their respective extensions in List 2.

LIST – 1:

1. Word file
2. Image file
3. Video file
4. Executable file

LIST – 2:

1. .mp4
2. .exe
3. .doc
4. .jpeg

1-C, 2-D, 3-B, 4-A

1-D, 2-B, 3-A, 4-C

1-D, 2-A, 3-C, 4-B

1-C, 2-D, 3-A, 4-B

Q26. Find the odd one out

Windows

Internet

Unix

Android

Q27.

Which of the following hardware components is typically a circuit board or a chip, installed on a computer so that it can connect to the network?

- NIC
- EC
- PCB
- Modem

Q28. Overheating of a microprocessor can be caused due to:

1. Improper connection of the processor fan
2. To overclock the CPU, jumpers may be configured
3. Incompatible voltage supply

Choose the correct answer from the options given below.

- Only 1 and 2
- Only 1 and 3
- All 1, 2 and 3
- Only 2 and 3

Q29. If you are using some special fonts in your Word document and want to embed them which of the following actions you should take?

- Protect the document
- Change the cache settings
- Change the Auto Recover time
- Preserve fidelity when sharing the document

Q30. Identify the types of networks that can be established by the following technologies.

1. **USB**
2. **Bluetooth**

Choose the correct answer from the options given below.

1. PAN
2. PAN

1. MAN
2. MAN

1. PAN
2. MAN

1. MAN
2. PAN

Answer Key & Solution

Section 1 - Aptitude

Q1

36

Solution

(T1+T2+T3)/3=37

(T2+T3+T4)/3=34

T4=T1*4/5

Subtracting T2+T3+T4=102 from T1+T2+T3=111 you find that T1-T4=9

Thus T1*5/5-T1*4/5=9, so T1=45 and T4=36

Q2

Rhombus

Solution

In BDC, Q and R are the mid-points of BD ad CD respectively.

QR || BC and QR = 1/2 BC.

similarly, PS ||BC and PS =1/2 BC

PS|| QR and PS = QR [each equal to `1/2 BC].

Similarly PQ || SR and PQ =SR [each equal to 1/2 AD]

PS =QR =PQ [AD =BC]

Hence, PQRS is a rhombus

Q3

3 22187/47710

Solution

$3^{X-2} = 5 \Rightarrow 3^{X} = 45 = 90/2$

$\Rightarrow X\log_{10}3 = \log_{10} 90 - \log_{10}2$

$= 2\log_{10}3 + 1 - \log_{10}2$

$= X(0.4771) = 1.65317$

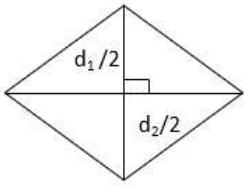
$\Rightarrow X = 165317/ 47710$

$= 3\ 22187/47710$

Q4

120

Solution



Area of rhombus = ½ ×product of diagonals

We know that, diagonals of a rhombus bisect each other at right angles.

Applying Pythagoras theorem:-

$(d_1/2)^2+(d_2/2)^2=side^2$

Where d1 and d2 are diagonals.

Given, measures of one side one diagonal of a rhombus are 13 cm and 24 cm.

$\Rightarrow 12^2+(d_2/2)^2=13^2$

$\Rightarrow (d_2/2)^2=25$

$\Rightarrow d_2 = 10$ cm

Area of rhombus = ½ × 24 × 10 = 120 cm^2

Q5

PQ min

Solution

The required answer is the LCM of P and Q. As P and Q are prime numbers, their LCM is their product.

Q6

8

Solution

—>$[\frac{15}{5}]$ ...= 3 (there are three zeroes at the end of 15!; 5^3)

—> $[\frac{15}{2}]+[\frac{15}{4}]+[\frac{15}{8}]+\ldots=7+3+1=11$ (—> 2^{11})

—> $[\frac{15}{3}]+[\frac{15}{9}]+\ldots=5+1=6$(—> 3^6)

—> $[\frac{15}{7}]+\ldots=2$ (—> 7^2)

So, $15!=2^{11}*3^6*5^3*7^2*11*13$

And eliminate the zeroes:
—> $\frac{15!}{1000}=2^8*3^6*7^2*11*13$

Now, just find the units digits and multiply one another to get the last digit —> ...6*...9...*9...*1...*3 = ...18...* 81 = ...8

The rightmost nonzero digit of **15! is 8**

Q7

534, 801

Solution

In a college of 1335 students, 60% are boys. Means we have to find 60% of 1335 to calculate number of boys.

= 60/100 of 1335

= 60 × (1335/100)

= 80100/100

= 801

Thus number of boys are 801 in the college

Now calculate the number of girls in the college as below

No. of girls = Total no of students - number of boys

No. of girls = 1335 - 801

No. of girls = 534

Thus, Number of boys is 801 and Number of girls is 534 in the college

Q8

64

Solution

2 White balls

3 Black balls

4 Red balls

Three balls are drawn

Total possible combination⇒$9P_3$

Combination without black⇒2C2×4C1+2C1×4C2+4C3

The combination with at least one black⇒9×8×73×2×1−[1×4+2×122+4]⇒12(7)−[20]⇒84−20=64

64 Combination includes atleast one black

Hence the correct answer is 64

Q9

12

Solution

Number of diagonals in a polygon can be found using , nC_2 - n

So, 54 = nC_2 - n

Solving this, we get n = 12

Q10

14/33

Solution

Number of ways of picking 2 chairs out of 12 = $12C_2$ = 66

Number of events of selecting 2 chairs for selecting no blue chairs = $8C_2$ = 28

So, the required probability = 28/66 = 14/33

Q11

Rs.150

Solution

Let the CP of the article be X

(50/100)*0.8 X=60

⇒ X = Rs. 150

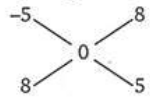
Hence, the cost price of the article is Rs. 150.

Q12

Rs. 70

Solution

By Allegation,



The ratio of price of pigeons sold at loss and profit respectively are in the ratio of 8: 5

∴ Cost price of profitable pigeon = $\frac{5}{8+5} \times 182 = ₹70.$

Q13

85

Solution

l = a+ (n-1)d

l = 7 + (n - 1)3

l = 3n + 4

Let us assume option (102), then

102 = 3n +4

n = 98/3 = 32 (2/3) = 32.66

$S_{32.66}$ = [(7 + 102)/2] x 32.66

= 1779.97

But $S_{32.66}$ = 1779.97 > 1242

Hence l must be less than 102

Let us assume option (79), then

79 = 3n +4

n = 25

Image not presentS_{25} = (7 + 79/2) x 25 = 1075

But since S_{25} = 1075 < 1242

Hence l must be greater than 79, therefore choice (85) is possible.

Let us consider choice (85).

L = 85 = 3 n + 4

=> n = 27

=> S_{27} = (7 + 85/2) x 27 = 1242

Hence (85) is correct.

Alternatively:

S_n = n/2 [2a + (n-1)d]

1242 = n/2[14 + (n -1)3]

=> n(3n + 11) = 2484

=> $3n^2$+ 11n – 2484 =0

=> $3n^2$ – 81n + 92n -2484 =0

=> 3n(n - 27) + 92 (n - 27)

=> n = 27 or n = -92/3

Hence n = 27 is the only admissible value.

=> l = a + (n-1)d

=7 + 26 x 3

=85

Q14

60

Solution

Let the two numbers be 3X and 4X

=> 3X + 15 : 4X + 15 = 4 : 5.

Solving, we get X = 15.

Hence the 2 numbers are 3X = 45 and 4X = 60. Hence, the bigger number is 60.

Q15

18 L

Solution

If x is the amount of milk present initially in the mixture, then we have

$(1 - 6/x)^3 = (8/27)$

solving this, we have

x = 18 liters

Q16

Rs5000

Solution

Difference between SI and CI for 3 yrs $=Pr^2/100^2*(3+r/100)$

$= Pr^2/100^3 (300 + r)$

$640=P*20^2/100^3*320$

$640=(P*20*20/100*100*100)*320$

P=Rs 5000

Q17

64 : 27

Solution

Let he deposits Rs. B in BOI and Rs. S in SBI.

Amount = $Principal\left\{(1 + Rate\%)^{time}\right\}$

Using the above formula,

Amount at BOI = $B (1 + 8\%)^3 = B (1.08)^3$

Amount at SBI = $S (1 + 44\%)^3 = S (1.44)^3$

After three years, the amounts become equal.

$\therefore B (1.08)^3 = S (1.44)^3$

$\Rightarrow \frac{B}{S} = \frac{(1.44)^3}{(1.08)^3} = \left(\frac{1.44}{1.08}\right)^3$

$\Rightarrow \frac{B}{S} = \left(\frac{4}{3}\right)^3 = \frac{64}{27}$

$\therefore B : S = 64 : 27$

Q18

12/5 days

Solution

1/C = 1/6 + 1/8

=> C = 24/7 days.

1/B + 1/C =

1/X =>1/8 + 7/24 =

> X = 12/5 days.

Q19 Darani, 250 m.

Solution

Total distance to be covered is 750 m.

Arun has a head start of 250 m. So, Distance to be covered by Arun and Darani = 500:750

Speeds are in the ratio of 1:3

So, the time taken will be 3:1

In 1 sec, if Darani covers 250 m, Arun takes 3 sec to cover the same distance.

So, to cover 750 m, Darani takes 3 sec.

But in that 3 sec, Arun would have covered 250 m out of the total 500 m to be covered. So, Darani wins the race by 250 m.

Q20 24

Solution

If a = 24, then polynomials will become

x^2-11x+24 and x^2-14x+48 which can be written as

(x-8)*(x-3) and (x-8)*(x-6)

(x-8) is common factor.

Q21 off

Solution

Call off-to decide that a planned event, especially a sports event, will not happen, or to end an activity because it is no longer useful or possible:

Q22 if both I and II are strong

Solution

Clearly, both the arguments in for and against are strong and enough. The conditions have to be agreed to, in order to save the life of the victims, though actually they ought not to be agreed to, as they encourage the sinister activities of the kidnappers.Hence both are strong.

Q23

DCBA

Solution

Statement 'D' gives proof of the popularity of the bowling alleys. Statement 'C' explains why they are popular. And 'B' followed by 'A' completes the series of statements leading to sentence '6'. DCBA is the correct option.

Q24

what's more

Solution

What's more-used to add something surprising or interesting to what you have just said

hence it is the right answer that suits the context of the sentence

Q25

love

Solution

Use 'love' here 'love-marriage' means 'to marry with beloved one'.

Q26

4

Solution

The error is in the part 4 of the sentence.

Generally, We don’t use nobody, no one, nothing, nowhere after no, not, never or other words which have a negative meaning (hardly, seldom). We use anyone, anybody, anything, anywhere.

Hence in the given sentence, instead of "no one" we use "anyone".

Q27

capital investment in commercial office buildings, even though vacancy rates in existing structures are exceptionally high and there is

Solution

Proper terminology used is capital investment and has been used only in options (capital investment in commercial office buildings, even though vacancy rates in existing structures are exceptionally high and there is) and (capital investment in commercial office buildings despite vacancy rates in existing structures that are exceptionally high, and although there is).

Hence options (capital to be invested in commercial office buildings even though there are exceptionally high vacancy rates high in existing structures with) and (investing capital in commercial office buildings even though the vacancy rates are exceptionally high in existing structures with) get rejected outright. Despite vacancy rates…. exceptionally high as in option (capital investment in commercial

office buildings despite vacancy rates in existing structures that are exceptionally high, and although there is) implies that the structures are exceptionally high whereas the original statement implied the rates as being high.

Hence option (capital investment in commercial office buildings despite vacancy rates in existing structures that are exceptionally high, and although there is) is also rejected and only option (capital investment in commercial office buildings, even though vacancy rates in existing structures are exceptionally high and there is) is the correct option.

Q28

The woman emphatically refused to confess guilt.

Solution

The statement said by the woman is strong and direct. refuse becomes refused. The answer is ' The woman emphatically refused to confers guilt.'

Q29

betrayal

Solution

Treachery means betrayal of trust.

Q30

thoughtful

Solution

Capricious means given to sudden and unaccountable changes of mood or behavior.

So its opposite will be predictable, predetermined or certain. Thus the close opposite is thoughtful.

Q31

Misologist

Solution

Misologist is a person who hates or distrusts reasoning, knowledge.

Q32

His diet program had been kept up for a month.

Solution

"had" become "had been" when changed to passive voice. keep up -move or progress at the same rate as someone or something else."He had kept up his diet program for a month." is the right answer

Q33

ETO

Solution

The first letter of every group is in alphabetical order, i.e. ABCDE Same is the case with II and III letters of the group. Such as PQRST and KLMNO

Q34

Mother

Solution

Since the man is referring to his nephew, the nephew's parent has to be a sibling to the man. Now the last part says that the man's sister has no other sister and hence the brother-sister relationship is established between the man and his nephew's mother. Now the lady in the picture is the grandmother of the nephew and hence she is the mother of the man's sister.

Q35

Sunday

Solution

1992 = 1600 + 300 + 92

Odd days in 1600 years = 0

Odd days in 300 years = (5 x 3)=1

92 = 23 leap + 69 ordinary years = 23*2 + 69 = 115 odd days = 3 odd days

No. of odd days till in 1993 till 14th March = 3 + 0 + 14 = 17 = 3 odd days

Odd days till 14th March 1993 = 1 + 3 + 3 = 7 = 0 odd days

So it was a Sunday.

Q36

120°

Solution

Speed of Minute hand = 360^0 / 60 = 6^0 / min

For 20 minutes it covers 120^0

Q37

4

Solution

From dice II and III 3 and 5 are two common numbers. Hence, 6 and 4 are opposite to each other.

Q38

I and III only

Solution

From I we know that either Girish or Prashant is the youngest.

From II we know Jayesh is the oldest, but it is irrelevant.

From III we know Girish is older than Prashant, so Prashant will be the youngest.

Therefore, we need only I and III.

order of oldest to youngest

J

K

G

P

Q39 if both the statements taken together are not sufficient and more information is required to solve the question

Solution

Statement (A) alone is not sufficient because it gives only the total salaries for Jai and Kishore.

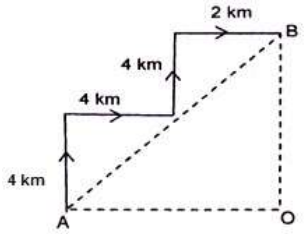
From statement (B) alone, it can be determined only that the sum of Jai's and Kishore's salaries in 1984 was Rs.55,000/1.1 = 50,000, which was given directly in statement (A).

Since statements (A) and (B) give the same information, taken together they still do not provide enough information, to determine Jai's salary in 1984.

Q40 10 km South-west

Solution

The path traveled by Amit is as follows.



Distance between A and B is $AB^2 = AO^2 + OB^2$

$AB = \sqrt{(8^2 + 6^2)}$

AB = 10 km

∴ Amit's home is 10 km from his office and is in south-west direction

Q41 218

Solution

The series is

3 + 69=72

72+57=129

129+44=173

173+30=203

203+15=218

The difference between the adding factors follows the below pattern

69 – 57 = 12

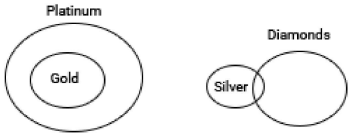
57 – 44 = 13

44-30 = 14

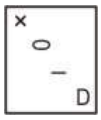
30-15 = 15

Q42 Only IV follows

Solution

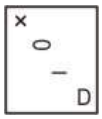


Only IV follows

Q43 

Solution

In one turn, the second and the third symbols move one step downwards; the fourth symbol becomes the second symbol and a new symbol appears at the uppermost position, in the next turn, the second and third symbols move one step upward; the first symbol becomes the third and a new symbol appears at the lowermost position. The process is repeated.



Q44. swept

Solution

No Solution

Q45. led

Solution

No Solution

Q46. instrumental

Solution

No Solution

Q47. to keep the seasons in their right months.

lution

he second sentence of para 3 renders option "to keep the seasons in their right

ıonths."to be the correct answer.

Q48. Julius Caesar.

Solution

The first sentence of the fourth para renders "Julius Caesar." to be correct.

Q49. an error of three days occurs after 10,000 years.

Solution

The last sentence of the passage makes "an error of three days occurs after 10,000 years.

" a suitable answer..

Q50. made a 10 day correction in the calculations.

Solution

The fourth sentence of the fourth para renders "made a 10 day correction in the

calculations." to be apt.

Q51. to add a day to make the calendar conform to the seasons.

Solution

The last two sentences of para 3, "In 120 years.... fourth year", renders option "to add a day

to make the calendar conform to the seasons." to be correct.

Q52. September

Solution

Profit = Sales - Cost.

In September, the profit is Rs.4 Lakhs and it is the highest.

Q53. 2.83 lakhs

Solution

Average Profit = 2+3+2+3+4+3 / 6 = 2.83 .

Q54. May

lution

ere we have to approximate the ratio of profit to the number of employees. It is lowest in

lay.

Q55. 5000

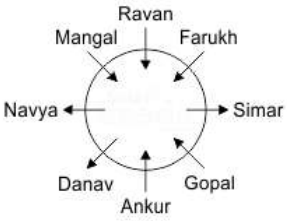
Solution

Number of employees in January is 11000 and the number in November is 16000. Hence 5000 more people were taken in by the company.

Q56. Mangal

Solution

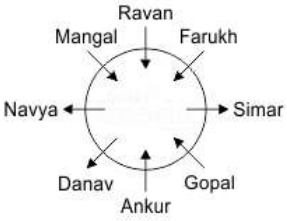
Mangal sits second to the right of Danav.



Q57. Mangal

Solution

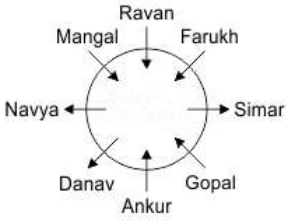
Mangal sits third to the left of Simar.



Q58. One

Solution

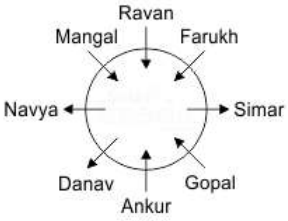
Only one person sits between Ravan and Navya when counted from right of Navya.



Q59. Ankur

Solution

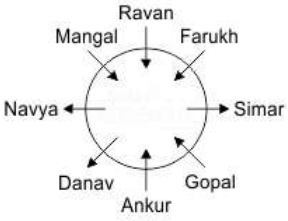
Ankur sits in front of Ravan.



Q60. Two

Solution

Two persons remain unchanged including Ankur.



Section 2 - Pseudocode and Computer Fundamentals

Q1 8

Solution

Initially a=3,b=2.

Inside first for loop a decremented thrice.(a=0)

Inside second for loop 'a' incremented by 2 thrice.(a=6)

Finally, 6+2 => 8 gets printed.

Q2 500

Solution

No Solution

Q3

30

Solution

No Solution

Q4

6

Solution

If block and else if block condition gets failed, so else gets executed and prints c value(6).

Q5

0

Solution

The arithmetic operator has higher priority than the Logical operator.So 6 / 23 = 0 (0 && anything = 0).

value of 'j' set to 0, 0+0 = 0 gets printed.

Q6

3

Solution

Inside for loop a decrement by 1(a=2) and breaks out of the loop because d< 2(d value set as 0 by default).

As a result it print 3 (2+1 = 3)

Q7

17 15 4

Solution

Apply VBODMAS rule to evaluate the expression.

For loop twice for a=1 and 2.

When a=1, b is updated as **17** and gets printed.

When a=2, b is updated as **15** and gets printed.

When a=3 loop condition gets failed and prints a+1 (i.e) **4**

Q8

test1 test completed

Solution

Apply VBODMAS rule to evaluate an expression.

6 / 7 = 0 so k value set as 0.

k lessthan equal to 0 (Ture) so executes nested if condition (i.e) k equals to 0 (true) so test1 gets printed and last instruction prints "test completed".

Q9

0

Solution

Initially a[4]= {0, 1,0,2}

a[0] = 0-0+2 (i.e){2,1,0,2}

a[1] = 2 -2 (i.e) {2,0,0,2}

a[2]= 0 - 0 (i.e) {2,0,0,2}

a[3] = a[2] (i.e) {2,0,0,0}

a[3] copied to m and prints m value 0.

Q10

it will print biography 1 time

Solution

Initially n=15

while condition true so "biography" gets printed and n=15+2 => n=17.

n value greater than 15 so loop terminates.

Q11

41

Solution

No Solution

Q12

888

Solution

No Solution

Q13

9

Solution

No Solution

Q14

2

Solution

The value of the variable a is 2 and b is 7, the first if condition becomes true so it will enter inside the if condition and checks the second condition if(b+1), it is true so it returns the value 2 from the function. Hence the output is 2.

Q15

A p t l

Solution

The characters in the even index will be printed. When an index is an odd number, space is printed.

So for given string "Alpital", **A p t l** gets printed.

Q16

Spam

Solution

Spam is digital junk mail: unsolicited communications sent in bulk over the internet or through any electronic messaging system.

Q17

3

Solution

Strong entities E1 and E2 are represented as separate tables. Then R1 one table and R2 requires one table. Totally 3 table.

Q18

1-B, 2-C, 3-A

Solution

IF function - Specifies a logical test to perform

NOT function- Reverses the logic of its argument

AND function - Returns TRUE if all of its arguments are TRUE

Q19

Cc

Solution

The CC (carbon copy) field allows you to send a copy of your email to multiple recipients.

Q20

Bcc

Solution

'Blind carbon copy' is a way of sending emails to multiple people without them knowing who else is receiving the email. Any emails in the

BCC field will be invisible to everyone else in the To and CC fields.

Q21

Google drive

Solution

Google Drive is a personal cloud storage service that lets users store giles and synchronize digital content across all devices.

Q22

1->Input device

2->Input device

3->Output device

4->Output device

Solution

Input devices: keyboard, joystick, mouse, light pen, track ball, scanner, graphite tablet, microphone, bar code reader.

Output devices: Monitor, Printer, Headphones, Computer Speakers, Projector, GPS, Sound card, Video card.

Q23

Function contains invalid numeric values

Solution

Excel shows this error when a formula or function contains numeric values that aren't valid. This often happens when you've entered a numeric value using a data type or a number format that's not supported in the argument section of the formula.

Q24

All 1, 2 and 3

Solution

Task can perform thru internet : Searching for Jobs, Shopping, Using a dictionary or other reference materials, download files/media, Getting directions, Banking, Investing/funding reports, Paying Bills, Checking the weather, Reading the news, Corresponding/sending email etc.

Q25

1-C, 2-D, 3-A, 4-B

Solution

Word file - .doc, .docm, .docx, .docx etc

Image file- .jpeg,.png,.gif,.tiff etc

Video file - .mp4, webm,mkv,flv etc

Executable file - .exe

Q26

Internet

Solution

All are OS except internet.

Q27

NIC

Solution

A network interface controller is a computer hardware component that connects a computer to a computer network. Early network interface controllers were commonly implemented on expansion cards that plugged into a computer bus.

Q28

All 1, 2 and 3

Solution

a) Processor fan may not be properly connected.
b) Heat sink may be not contacted with the processor.
c) Jumpers may be configured to over clock the CPU.
d) Voltage supply incompatible

Q29

Preserve fidelity when sharing the document

Solution

Preserve fidelity used to preserve the look of the document when you share the document with someone who uses a different version of Word.

Q30

1. PAN
2. PAN

Solution

Wireless Personal Area Network is developed by simply using wireless technologies such as WiFi, Bluetooth. It is a low range network. Bluetooth is a personal area network (PAN) standard that is lower power than 802.11. It was originally specified to serve applications such as data transfer from personal computers to peripheral devices such as cell phones or personal digital assistants.